

DONETSK, Ukraine

WMO: 345190

Lat: 48.072N

Lon: 37.727E

Elev: 225

StdP: 98.65

Time Zone: 2.00 (EUE)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.0	-22.2	0.5	-18.0	-19.3	0.7	-15.3	13.8	-5.8	12.3	-7.6	3.4	70	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.6	19.2	31.6	19.0	29.8	18.7	21.4	28.5	20.5	27.5	19.7	26.7	4.6	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.3	23.8	18.2	13.4	22.8	17.4	12.8	22.4	63.2	28.4	59.9	27.3	57.3	26.7	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.0	8.7	DB	-22.4	35.2	3.8	2.4	-25.1	36.9	-27.4	38.3	-29.5	39.6	-32.2	41.3	(4)
			WB	-22.7	23.1	3.7	1.3	-25.4	24.0	-27.6	24.8	-29.7	25.5	-32.4	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	-4.2	-3.8	1.7	9.8	16.0	19.8	22.5	21.5	15.6	9.0	2.6	-2.6	(6)
(7)		DBStd	10.73	5.91	5.95	4.69	4.17	4.56	3.69	3.42	3.89	4.19	4.87	4.96	5.58	(7)
(8)		HDD10.0	1846	441	386	261	52	6	0	0	0	6	77	226	392	(8)
(9)		HDD18.3	3748	699	619	517	255	100	27	4	12	100	292	472	650	(9)
(10)		CDD10.0	1501	0	0	2	47	193	293	386	357	174	46	3	0	(10)
(11)		CDD18.3	361	0	0	0	1	29	69	132	111	18	2	0	0	(11)
(12)		CDH23.3	3503	0	0	0	9	268	635	1316	1126	138	11	0	0	(12)
(13)		CDH26.7	1286	0	0	0	1	63	202	529	466	23	2	0	0	(13)
(14)	Wind	WSAvg	4.1	4.7	5.0	5.0	4.3	3.6	3.4	3.3	3.5	3.7	4.0	4.3	4.8	(14)
(15)	Precipitation	PrecAvg	517	39	33	38	39	45	71	49	38	42	39	38	44	(15)
(16)		PrecMax	758	115	100	77	106	89	181	192	118	126	90	88	90	(16)
(17)		PrecMin	348	11	8	15	4	1	16	5	0	1	10	6	13	(17)
(18)		PrecStd	100	23	21	17	20	23	40	40	30	33	19	23	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	9.8	16.9	24.7	31.1	33.1	35.8	36.8	29.2	24.8	16.0	10.1	(19)
(20)		MCWB	6.9	7.6	10.1	14.3	17.5	19.5	19.6	19.3	17.0	16.1	11.8	8.8	(20)	
(21)		2%	DB	5.9	6.1	13.4	21.1	28.2	30.9	33.8	33.9	26.8	20.9	13.0	7.8	(21)
(22)		MCWB	5.2	4.8	8.2	12.4	16.5	19.1	19.6	18.6	16.7	14.1	10.7	7.2	(22)	
(23)		5%	DB	4.1	4.2	10.7	19.0	26.2	29.0	31.8	31.2	24.9	18.1	10.9	5.8	(23)
(24)		MCWB	3.6	3.3	6.6	11.3	16.0	18.7	19.5	18.6	15.6	13.6	9.1	5.2	(24)	
(25)		10%	DB	2.6	2.9	8.2	17.1	24.2	27.1	29.9	29.0	22.9	16.0	9.1	4.1	(25)
(26)		MCWB	2.1	2.2	5.5	10.4	15.6	18.0	19.2	18.1	15.1	12.6	7.9	3.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	7.6	10.6	14.8	19.3	22.1	23.1	21.8	20.3	17.4	12.7	9.2	(27)
(28)		MCDWB	8.0	9.5	15.5	22.7	25.6	28.8	30.2	29.9	26.5	21.8	14.5	9.8	(28)	
(29)		2%	WB	5.2	5.0	8.9	13.4	18.2	20.6	21.7	20.6	18.5	15.6	10.9	6.9	(29)
(30)		MCDWB	5.6	5.6	12.6	19.2	25.3	27.2	28.8	28.3	23.4	19.1	12.7	7.5	(30)	
(31)		5%	WB	3.6	3.5	7.2	12.3	17.2	19.8	20.9	19.8	17.4	14.3	9.4	5.1	(31)
(32)		MCDWB	4.0	4.1	9.9	17.3	23.9	26.5	27.8	27.1	22.0	17.4	10.6	5.6	(32)	
(33)		10%	WB	2.0	2.2	5.7	11.2	16.2	18.9	20.0	19.1	16.4	12.9	7.9	3.6	(33)
(34)		MCDWB	2.5	2.7	7.8	15.7	22.7	25.2	27.1	26.3	20.6	15.5	9.0	4.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	5.1	5.8	7.2	9.6	10.6	10.6	11.1	11.5	10.3	7.9	5.7	4.8	(35)	
(36)		5% DB	MCDBR	4.9	6.1	11.3	12.3	12.6	12.9	13.6	12.3	10.5	8.0	5.6	(36)	
(37)		MCWBR	4.2	5.1	6.9	6.0	5.3	4.9	4.6	4.9	5.6	5.9	5.9	4.9	(37)	
(38)		5% WB	MCDBR	4.7	5.6	9.9	10.7	11.6	11.1	11.3	11.7	10.3	9.4	7.0	5.1	(38)
(39)		MCWBR	4.3	4.9	6.7	5.8	5.7	5.1	5.1	5.1	5.5	6.1	5.7	4.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.356	0.391	0.449	0.412	0.412	0.437	0.435	0.407	0.373	0.348	0.331	(40)	
(41)		taud	2.210	2.141	2.175	2.077	2.270	2.295	2.231	2.225	2.274	2.338	2.320	2.258	(41)	
(42)		Ebn at Noon	713	775	813	799	852	852	823	806	791	757	691	659	(42)	
(43)		Edh at Noon	81	109	124	151	129	127	133	129	111	89	74	70	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.96	1.89	2.96	4.44	5.86	6.24	6.18	5.60	3.94	2.31	1.08	0.68	(44)	
(45)		RadStd	0.20	0.37	0.40	0.45	0.52	0.52	0.38	0.39	0.39	0.35	0.20	0.11	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.94	N/A	N/A	N/A	-0.65	N/A	N/A	-277	+195	+105	(46)
(47)	Regional (9 neighbors)	+0.70	N/A	N/A	+1.12	N/A	N/A	N/A	-170	+118	+80	(47)

Nomenclature: See separate page